Fake News Classifier with Deep Learning

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***Abstract*—In this decade, social media has become more frequently used because it is easy to see some news on the social media for example Instagram, Facebook, and other social media. However, the problem that often arises in social media problems is the accuracy of the news. There are many fake news circulating on social media. This paper aims to classify the fake news using random forest. This research uses data Kaggle "Fake News Dataset Combined Different Sources". The accuracy of the Fake News Classifier using this model is is 0.9403, the Classification Report on this model is True Recall Score : 0.88 and the Fake Recall score is 0.98 while the Precision Score is 0.96 and 0.93 for each true and fake result , F1-Score is 0.92 and 0.95. At the end the model using self-inputted data is tested**

Keywords— Fake News Classifier, NLP, Deep Learning, Random Forest Classifier

I. INTRODUCTION

Fake news is false or misleading information presented from news that could give people some misunderstanding about a situation or information [1] [2]. Fake news can reduce the impact of real news by competing with it. Indonesia 60% of internet users believe in fake news. The number of people who believe in fake news increases every year [3][4].

In 2021, almost all peoples use social media, sometimes they post news to social media because of that the percentage of fake news increased especially on social media and some people don't know if that news is true or not . Ironically, sometimes website news also contains fake news [5][6].

This paper focuses making a model that can detect the classes of some news is it fake or not. On processing the text we vectorize the text by using TF-IDF. TF-IDF stands for "Term Frequency-Inverse Document Frequency" ; This is a technique for processing the text into numbers. Generally this technique computes a score to each word to give its importance in the document and corpus, TF-IDF usually used for Information Retrieval or Text Mining [7][8][9][10] .Random Forest Classifier is used to classify the news classes after being vectorized. Random Forest is good for classify text [11] [12][13].

II. RELATED WORK

Fake News is closely associated with politics, and can unhelpfully narrow the focus of the issue [3]. Lot of things read online especially in the social media feeds may appear to be true, but often not. Usually the story is created to either influence people to view the news , push a political agenda and cause confusion even sometimes to be a profitable business for online publishers. Fake news can deceive people by looking like a trusted website or using a similar name and web address to the real ones.

Fake news has become a hot topic since 2017, because the internet has enabled a new way to publish, share and consume information. Therefore many people can get news from social media sites and networks and it can be difficult to tell whether the information is fake or not. There are different types of fake news like parody, misleading news that is sort of true but used the wrong context, sloppy reporting that fits an agenda, and misleading news that is not based on fact .

TF-IDF short of Term Frequency-Inverse Document Frequency, is a numerical static that is intended to reflect how important a word is to a document. Tf-IDF is often used as a weighting factor in searches of information retrieval, text mining, and user modeling. A survey in 2015 showed that 83% of text based recommender systems in digital libraries use TF-IDF [14].

TF-IDF have 2 components TF(Term Frequency) which measures the frequency of a word in a document

$$Tf = \frac{Nword}{Tword} \quad (1)$$

Eq. (1) is an equation for TF which *Nword* is the frequency of a word on a document and *Twords* is the total of words on document. For IDF(Inverse Document Frequency) which measures the rank of the specific word for its relevance within the text as shown in Eq. (2).

$$IDF = Log\left(\frac{Ndoc}{Ndoc\,t'}\right) \quad (1)$$

Ndoc is a number of documents and *Ndoc t'* is number of documents that have a specific word. And for TF-IDF which combine TF and IDF as shown Eq. (3)

$$Tf\text{-}idf = Tf \,.\, idf \quad (3)$$

Eq. (3) is an equation for combining the TF and IDF in order to gain the value of TF-IDF of a specific word .

Random Forest is a classifier that contains a number of decision trees on various subsets of a given dataset. The determination of the random forest classification was taken based on the voting results of the trees formed. The

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winner is determined by the most votes from the tree formed. The flow of random forest can be seen in Figure 1

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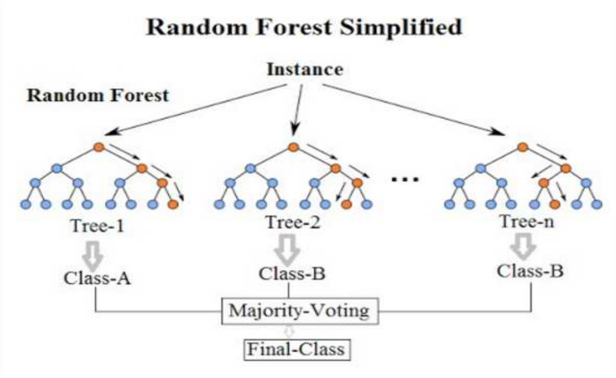

Fig 1. Random Forest Classification

In a normal decision tree, when splitting a node, one of every possible feature is picked that produces the most separation between the observations in the left node. In contrast, each tree in a random forest can pick only from a random subset of features. This forces even more variation amongst the trees in the model and ultimately results in lower correlation across trees and more diversification The Random Forest model does not offer as much interpret ability as a single tree and the performance is a lot better.

III. Proposed Method

This project uses python as programming language and utilizes google colab to run our code for its cloud computing. This model utilized the keras library. This library is used to assist in the work process. The sklearn library imports the Random Forest Classifier. The TfidfVectorizer used in data pre-processing, seaborn is also used to map what the plotting and matplotlib.pyplot to plot it along with sklearn.metrics to evaluate the result of our training. The dataset takes from Kaggle with entitled “Fake News Dataset Combined Different Sources” [15]. There are 69.000 rows of data that consist of id (Unnamed), title, text, and Ground label. It is used to train and test the accuracy of the proposed classifier.

However, there some data contains null values in a title and text as shown Fig. 2. This data is dropped since it is only 1 % of all data. Moreover the title column is dropped.

Null Value

Column	Null Count
Title	680
Text	67
Ground Label	0

Fig 2. Null Detected

Before the text is tokenized, the text is pre-processed, it is to eliminate the stop words in the dataset and tokenize it then eliminate punctuation using regex as shown in Figure 3. To ensure the model can be tested after training it, the dataset is split into 47808 rows of data and 20490 testing data or 30% of the datas for testing and the rest of it for training.

```
def wordopt(text):
    text = text.lower()
    text = re.sub('\[.*?\]', '', text)
    text = re.sub("\\W"," ",text)
    text = re.sub('https?://\S+|www\.\S+', '', text)
    text = re.sub('<.*?>+', '', text)
    text = re.sub('[%s]' % re.escape(string.punctuation), '', text)
    text = re.sub('\n', '', text)
    text = re.sub('\w*\d\w*', '', text)
    return text
```

Fig. 3 Regex Function

Finally, before the classifier is trained, The TfidfVectorizer() is used to change the dataset from string to numerical that the model can train in as shown in Figure 10 and that finishes the preprocessing process. After the preprocessing process finish, the model is trained

IV. Analysis Result

From several explorations we did in this paper, we found out that the title is not crucial in fake news classification so we dropped it. Through the score function provided in the classifier we obtained the model accuracy based on our test dataset that consists of 20490 testing data.

The accuracy of the model produced based on the test dataset is 93.67% accuracy and at maximum can reach 95%.

	precision	recall	f1-score	support
fake	0.93	0.97	0.95	12414
true	0.96	0.88	0.92	8076
accuracy			0.94	20490
macro avg	0.94	0.93	0.93	20490
weighted avg	0.94	0.94	0.94	20490

Fig. 4 Classification Performance

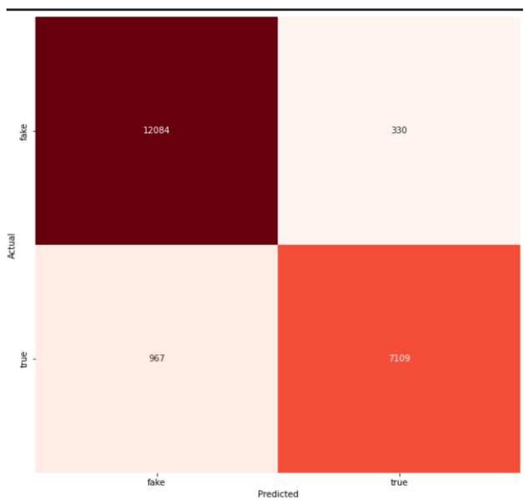

Fig 5 Confusion Matrix

To further give more insight into the result of the test on the model, the classification performance is presented on Fig.4 and the confusion matrix on Fig. 5.

The accuracy of the classification model reached a testing accuracy of 93.67% and maximum can reach 95% in other using the same test data. On this data the model can flawlessly detect the fake news from the Fake News dataset. The model is also tested by trying to make a live input to the model and make the model predict it. The custom input is obtained from a news website.

Text = "Amazon notched up a legal win on Friday, as a Luxembourg court suspended a €746,000 fine the US retailer had to pay each day as part of its fight with the Grand Duchy data protection regulator over suspected data privacy breaches. Luxembourgs National Commission for Data Protection (CNPD) hit Amazon with a record €746 million fine in July, saying that 'Amazon's processing of personal data did not comply with the EU General Data Protection Regulation,' (GDPR), according to a regulatory filing by Amazon in the US.On top of the fine, which Amazon is currently appealing, the CNPD also ordered the company - whose European headquarters are located in Luxembourg - to pay €746,000 a day until it had rectified the gaps noted by the regulator. But the president of Luxembourg's Administrative Court on Friday ruled to suspend the daily payments, saying the watchdog's instructions on how to correct the breaches were too vague, the court said in a press release. Luxembourg's court will next year rule on Amazon's appeal to pay the €746 million fine, with a number of hearings between the two parties still ongoing. GDPR legislation empowers national data regulators to levy penalties of as much as 4% of a company's annual revenue"

Prediction: This is not a Fake News

Fig 6 Example data with Prediction "Not Fake"

Figure 6 is one of the examples to input the real news from Amazon.com. The model correctly predicts the result as not fake news because the news from Amazon.com have been verified to be not fake news.

Figure 7 is the data that has been labeled fake from the dataset because there is lack of information of websites that usually gave a fake news, therefore the model can only be tested using the data from the dataset.

Text = "It s hard to believe, but Donald Trump does have a sizable amount of supporters who agree with his ideas on immigration and guns. Unsurprisingly, one of those people is Ann Coulter. A racist in her own regard, she falls perfectly into Trump s fan base.After the shooting massacre at a gay nightclub in Orlando, Florida, Coulter tweeted out: To my gay friends: Please consider the possibility that Hillary s immigration policies might get you killed. See Trump s speech today. To my gay friends: Please consider the possibility that Hillary's immigration policies might get you killed. See Trump's speech today. Ann Coulter (@AnnCoulter) June 13, 2016Okay, first, what gay person would be her friend? And before you get offended in any regard to that question (you know who you are) I m a lesbian and find Coulter repulsive on every level. Which leads to the bigger question what person, in general, would be her friend?During his speech, Trump of course spent time doubling-down on his racist and Islamophobic remarks that Muslims should be banned from immigrating to the United States.Does Coulter (and Trump) not even recognize the fact that the shooter (whose name I ll never mention, because that s what he wanted) was actually born in the United States, and was able to access his arsenal of weaponry and ammunition because of Florida s, and the United States, lax gun laws? Just like almost every shooter before him, and the only way to truly combat this sort of issue is to try to prevent them from happening in the first place.You can t always change a mindset, but you sure as hell can try to prevent a potential terrorist from owning a gun.The thing is, this is just Ann Coulter being Ann Coulter. She s a bigot and she s proud to be a bigot, and she doesn t care if you think she s a bigot. She s an isolationist who seems to think white people should reign supreme.It s just amusing to think that she wants us to believe she has gay friends. If, however, she really does have gay friends, they should probably realize they re friends with a bigot.Featured Photo by Frederick M. Brown/Getty Images Twitter"

Prediction: This is a Fake News

Fig 7 Example data with Prediction "Fake"

V. CONCLUSION

This paper can prove classifying the fakenews using random forest. However the model trained by giving the model a past news therefore this model can only detect the past news not the today news. To improve this project the collection data could make a real time news detection by using web crawling to some of websites that have already verified trusted websites using other machine learning and fake news from other websites by searching their similarity to their title. By using these methods we can get the data real time and could detect the current news. This project also only used the English version. It could be improved by making the model that could input other languages so it can detect news from different countries.

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